📊 Customer Churn Analysis

🔍 Project Overview

Customer churn is a critical business problem where customers stop using a company's service. This project aims to analyze customer data to identify patterns, trends, and key factors that influence churn.

Understanding these patterns helps businesses:

- Improve customer retention
- Reduce revenue loss
- Make data-driven decisions

🎯 Objective

The main objectives of this project are:

- Analyze customer behavior and churn patterns
- Identify key features affecting churn
- Perform **Exploratory Data Analysis (EDA)**
- Generate actionable business insights

📂 Dataset Description

The dataset contains customer-level information with the following features:

Feature	Description
Gender	Male/Female
SeniorCitizen	Whether customer is senior (0/1)
Tenure	Number of months customer stayed
Contract	Type of contract (Monthly, Yearly)
MonthlyCharges	Monthly bill amount
TotalCharges	Total amount charged
Churn	Whether customer left (Yes/No)

⚙️ Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

🧹 Data Cleaning & Preprocessing

Real-world datasets are often messy. Before analysis, the following steps were performed:

✓ Handling Missing Values

- TotalCharges column had blank values
- Replaced missing values with 0

👉 *Why?* Missing data can distort analysis and lead to incorrect conclusions.

✓ **Data Type Conversion**

- Converted TotalCharges from **object (string)** → **float**

👉 *Why?* Numerical operations (mean, sum, etc.) require numeric data types.

✓ **Feature Transformation**

- Converted SeniorCitizen from 0/1 → No/Yes

👉 *Why?* Improves readability and makes visualizations clearer.

📊 **Exploratory Data Analysis (EDA)**

EDA is the most important step where we explore the data visually and statistically.

📌 **1. Churn Distribution**

- Used **count plot** and **pie chart**
- Helped understand overall churn rate

👉 Insight:

- Majority of customers stayed, but churn rate is still significant

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📌 **2. Churn by Gender**

- Compared churn across male and female customers

👉 Insight:

- No strong difference — gender is not a major factor

📌 **3. Churn by Senior Citizen**

- Compared churn between senior and non-senior customers

👉 Insight:

- Senior citizens show slightly higher churn behavior

📌 **4. Churn by Tenure**

- Used **histogram**

👉 Insight:

- Customers with low tenure (new users) churn more
- Long-term customers are more loyal

📌 **5. Churn by Contract Type**

- Compared:
  - Month-to-month
  - One-year
  - Two-year contracts

👉 Insight:

- **Month-to-month customers churn the most**
- Long-term contracts reduce churn

📌 **6. Charges Analysis**

- Analyzed **MonthlyCharges vs Churn**

👉 Insight:

- Customers with higher monthly charges tend to churn more

📈 **Key Insights (Business Perspective)**

- 📉 Short-term contracts increase churn risk
- 📈 Long-term customers are more stable
- 💰 Higher charges may lead to dissatisfaction
- 👴 Senior citizens require targeted retention strategies

🧠 **Beginner Concepts Explained**

🔷 **What is Churn?**

Churn refers to customers who stop using a service.

🔷 **What is EDA?**

EDA (Exploratory Data Analysis) means:

- Understanding data
- Finding patterns
- Detecting relationships

🔷 **What is a Count Plot?**

A count plot shows the number of occurrences of each category.

Example:

- Number of churned vs non-churned customers

🔷 **What is a Histogram?**

A histogram shows the distribution of numerical data.

Example:

- Distribution of customer tenure

🔷 **What is Correlation?**

Correlation measures how strongly two variables are related.

- Positive → both increase together
- Negative → one increases, other decreases

🔷 **What is GroupBy?**

groupby() is used to:

- Split data into groups
- Perform operations like mean, count, sum

🚀 **Conclusion**

This project demonstrates how data analysis can uncover valuable insights into customer behavior. By identifying churn patterns, businesses can take proactive steps to improve retention and customer satisfaction.

🔮 **Future Scope**

- Build a **Machine Learning model** for churn prediction
- Deploy a dashboard using **Power BI** or **Tableau**
- Perform feature engineering for better insights
- Use advanced techniques like clustering

📌 **Project Learnings**

- Data cleaning is crucial before analysis
- Visualization helps in better understanding
- Business insights are as important as technical work